MILITARY SPECIFICATION

BODY ASSEMBLY, TANK PERISCOPE: 10513620

1. SCOPE

1.1 This specification covers one type of periscope body assembly which receives a reflected line of sight from a head assembly to form a tank periscope (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

F10513620	Body Assembly

PACKAGING DATA SHEET

MIL-P-14232/P10513620	Packaging of Body Assembly: 10513620

FSC 6650

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The body assembly shall be manufactured in accordance with Drawing F10513620 and drawings pertaining thereto.

3.2 General specification. - The contractor shall be responsible for adherence to and compliance with the following requirements of MIL-F-13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

3.3 Environmental.

3.3.1 Storage temperatures. - The body assembly shall show no evidence of physical failure such as glass breakage, bonding failure, or casting failure due to storage at temperatures ranging from -80° to +160° F.

3.3.2 Rigidity. - The following requirements shall be met during and subsequent to being vibrated at 30 cycles per second and an amplitude of 0.06 inch (0.12 inch total excursion). The catches shall be capable of securing the body assembly rigidly (no apparent separation) against the locating surfaces of head and mount with which it is intended to be used.

3.3.3 Catches (operability). - The two catches shall be manually operable at temperatures ranging from -40° to +150° F.

3.4 Cleanliness. - The body assembly shall meet the cleanliness requirements of MIL-O-13830 and there shall be no more than four particles of dirt visible on the optical surfaces. The area of each particle shall not be in excess of an area equivalent to a size ten dig.

3.5 Sealing. - The body assembly shall withstand an internal pressure of 5 ± 0.10 pounds per square inch gage (psig) with no evidence of leakage.

3.6 Light deviation. - An incident line of sight, which is perpendicular to a plane defined by the machined surfaces of the four pads, shall emerge parallel to that plane within 45 minutes of arc. The incident line of sight shall also emerge parallel to a plane, defined by the 3/4" x 3/4" with 3/8" radius surface adjacent to Valve Assembly 8201751, within 45 minutes of arc.

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3.7 Image tilt. - The image of a target line, originally perpendicular to a plane, defined by the 3/4" x 3/4" with 3/8" radius surface adjacent to Valve Assembly 8201751, shall emerge perpendicular to the same plane within 45 minutes of arc.

3.8 Spherical power. - Spherical power shall not exceed 0.10 diopter when scanning the exit aperture with a telescope having a 1.0 inch aperture.

3.9 Astigmatism. - Astigmatism shall not exceed 0.10 diopter when scanning the exit aperture with a telescope having a 1.0 inch aperture.

3.10 Interior atmosphere. - The interior atmosphere of the body assembly shall contain nitrogen having a dew point at least as low as low -25° F.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection of components and sub-assemblies not covered by specifications or Supplementary Quality Assurance Provisions (SQAP) shall be in accordance with 4.2.2. Definitions of inspection terms not otherwise defined herein shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot. - Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examination and tests. - Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105, and the inspection level and sampling plans specified in Table I titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

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TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of MIL-STD-105

	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0%		
101. Rigidity	3.3.2	4.5.2.2
102. Cleanliness	3.4	4.5.3
103. Light deviation	3.6	4.5.5
104. Image tilt	3.7	4.5.6
105. Spherical power	3.8	4.5.7
106. Astigmatism	3.9	4.5.8
MINOR: NONE		

4.2.3 Disposition of non-conforming product. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 Control tests.

4.3.1 General. - One body assembly shall be selected at random as a control sample from each 50 produced, or from each month's production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Visual and Drawings
302. General specifications	3.2	MIL-F-13926

4.3.2 Environmental and functional. - Three body assemblies shall be selected at random as a control sample from each 50 produced, or from each month's production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
303. Storage temperature	3.3.1	4.5.2.1
304. Operability	3.3.3	4.5.2.1
305. Sealing	3.5	4.5.4
306. Interior atmosphere	3.10	4.5.9

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4.3.3 Control test failure. - Should any one item of a control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall maintain all required measuring and testing equipment in accordance with the applicable requirements specified in MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" requirements of MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.2 Devices used for environmental testing. - All environmental testing devices used to determine conformance of the environmental requirements listed below shall conform to the "Test facilities" requirements outlined in MIL-F-13926:

(a) Temperature (3.3.1, 3.3.3)
(b) Rigidity (3.3.2)
(c) Sealing (3.5)
(d) Interior atmosphere (3.10)

4.5 Test methods and procedures.

4.5.1 Fabrication. - The body assembly shall be subjected to a visual, tactile and dimensional inspection to determine compliance with 3.1 and the applicable drawings and requirements covered by this specification. Dimensional inspection shall be performed using standard measuring equipment.

4.5.2 Environmental.

4.5.2.1 Storage and operability temperatures. - This test shall be conducted in accordance with Procedure I of the temperature cycling test in MIL-F-13926 at the temperatures specified in 3.3.1 and 3.3.3. The body assembly shall be

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mounted on a device so constructed as to permit clamping of the body assembly securely to the device when the catches reach their apex and are fully closed. While exposed to the temperature extremes specified in 3.3.3 the catches shall be manually operated to determine compliance with the requirements of 3.3.3. Upon completion of the specified tests and return to standard ambient temperature, (60° to 90° F) the body assembly shall be inspected for evidence of physical damage such as casting failure or glass breakage. It shall then be subjected to the tests specified in 4.5.2.2 to 4.5.9 inclusive to determine conformance with the requirements referenced therein.

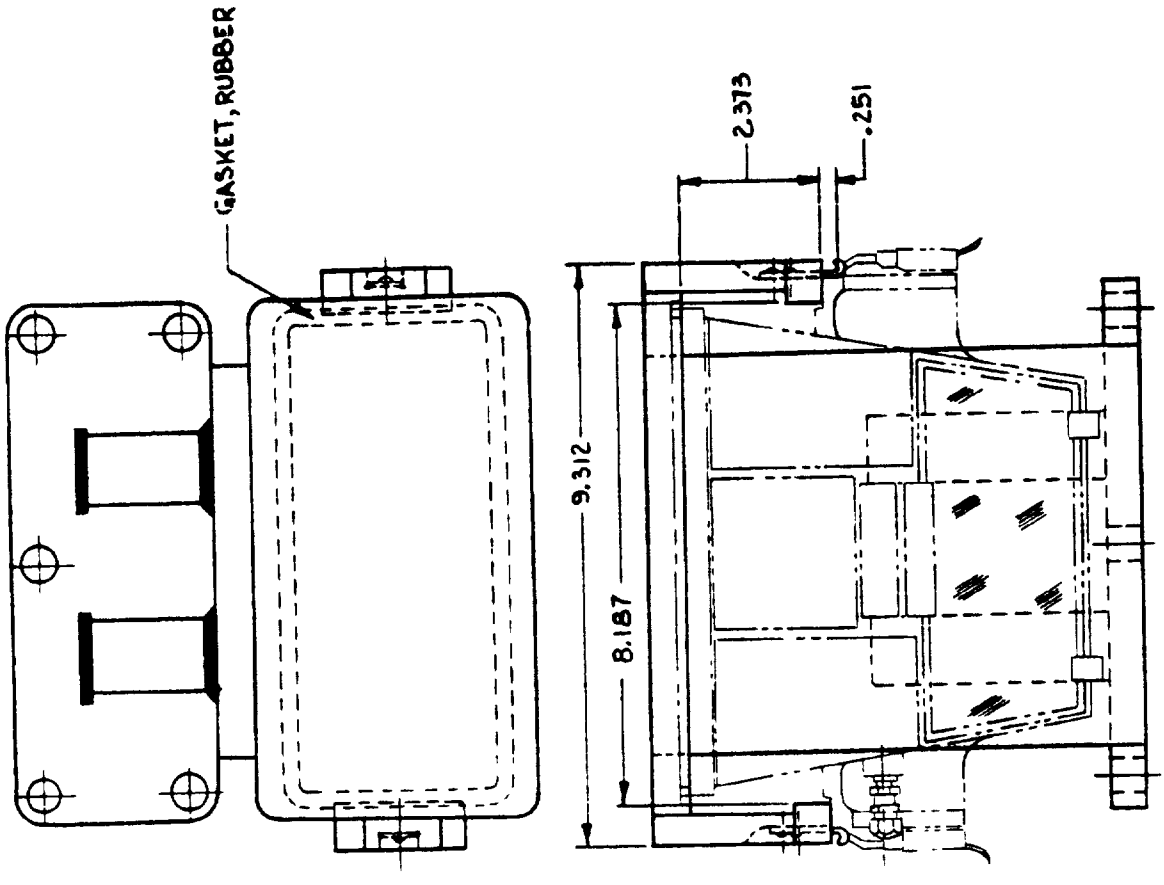
4.5.2.2 Rigidity. - This test shall be accomplished by means of a test fixture as shown in Figure 1. The fixture shall be clamped to a vibration table. Clamp the body assembly in the fixture by means of the catches and immobilize catches to prevent accidental opening during vibration. The body assembly shall be vibrated in a vertical plane at the amplitude and frequency specified in 3.3.2 for a period of 2½ minutes ± 15 seconds. The vibration amplitude shall be measured at the extreme bottom portion of the body assembly as positioned in the fixture. During vibration, observe rigidity of the body assembly in the test fixture using a stroboscope to produce the optical effect of stopping the vibration motion to determine rigidity at the locating and mating surfaces. The body assembly shall remain secure against the test fixture receiving surfaces to insure compliance with the requirements of 3.3.2. Subsequent to vibration the body assembly shall be subjected to a visual and tractile examination and shall show no evidence of physical damage or failure. The body assembly shall then be subjected to the tests specified in 4.5.3 to 4.5.9 inclusive to determine compliance with the requirements referenced therein.

4.5.3 Cleanliness. - Optical surfaces of the completed body assembly shall be inspected for cleanliness in accordance with MIL-O-13830 and shall meet the requirements specified in 3.4. The mean diameter of a size ten dig is equal to 0.10 millimeter, when determining sizes of dirt particles.

4.5.4 Sealing. - Standard pressure testing equipment capable of measuring and monitoring the internal pressure of the body assembly shall be utilized for the sealing test. Pressurize the body assembly with dry nitrogen gas to the psig value specified in 3.5 for a period of 5 minutes. During this time period the interior pressure shall not decrease in compliance with the requirement specified in 3.5.

4.5.5 Light deviation. - This test shall be performed utilizing an auxiliary telescope of at least 3 power, containing a horizontal and vertical reticle line, mounted on a supporting device capable of holding the telescope and body assembly in a position permitting an incident line of sight to be directed from a calibrated target in a collimator projector, through the body assembly to the auxiliary telescope. Utilize the space in between the pads and the back surface of the 3/4 inch keyways to centrally position the body assembly in the device with the machined top surfaces of the pads on the same plane with the line of sight through the telescope. The target pattern shall consist of a horizontal and vertical cross line with limit marks 45 minutes from the intersection of the crossline. Limit marks shall also be added in each quadrant, offset 45 minutes of arc from the horizontal crossline, to subsequently measure image tilt. The collimating projector shall be positioned in front of the body assembly with the center

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RIGIDITY HOLDING FIXTURE

Figure 1

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vertical line of the target in a plane parallel to a plumb line and on the same plane with the back surface of the 3/4 inch keyway adjacent to the valve assembly. The auxiliary telescope reticle line shall be adjusted parallel to a plumb line. With the body assembly installed in the device, the line of sight shall be determined by viewing through the telescope and observing coincidence between the intersecting point of the auxiliary telescopes reticle and the corresponding intersecting point of the collimator target line. This coincidence must be maintained within the limits of the tolerance line of the collimator target to assure conformance with the requirement outlined in 3.6.

4.5.6 Image tilt. - This test shall be performed utilizing the same test set-up and target specified in 4.5.5. When an incident line of sight is directed from the target, through the body assembly to the auxiliary telescope, the image of the target vertical line shall emerge parallel to the vertical line of the auxiliary telescopes reticle within the limits of the tolerance line of the collimator representing image tilt to conform with the requirement of 3.7.

4.5.7 Spherical power. - This test shall be performed utilizing a four line resolving power chart as shown in Figure I of MIL-O-13830 with the test set up specified in 4.5.5 except that a millidioptometer of 20 power with an objective aperture of 1.0 inch shall be used in place of the auxiliary telescope. The distance between the millidioptometer objective and the test chart construction shall be in accordance with the "Resolution test" specified in MIL-O-13830. The millidiopter scale shall be set to zero diopters after best focus of the target lines has been obtained. Position the periscope body and supporting device between the millidioptometer and test target and refocus the millidioptometer to obtain a best image of the target line. The difference in scale reading is to be considered spherical power. This difference shall not exceed the limits outlined in location over the face of the prism.

4.5.8 Astigmatism. - This test shall be performed using the set up specified in 4.5.7. The image of the target lines in the four meridians shall be in sharp focus when viewed with the millidioptometer within the adjustment specified in 3.9. Repeat this test at a minimum of five different locations over the face of the prism.

4.5.9 Interior atmosphere. - Upon satisfactory completion of the preceding tests, the body assembly shall be flushed with dry nitrogen gas as specified in 3.10 and sealed. The flushing shall continue until the gas bled from the body assembly has a dew point at least as low as minus 25 degrees fahrenheit when checked by suitable measuring equipment.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10513620. The level of protection shall be as specified in the procurement document.

6. NOTES

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6.1 Intended use. - The Tank Periscope Body Assembly 10513620 is used together with the Tank Periscope Head Assembly 10513625 to form the XM47 Tank Periscope. The XM47 Tank Periscope is primarily intended for use as a driver's daylight periscope in the XM551 Vehicle.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable level of preservation, packaging and packing
(c) Applicable stock number
(d) Applicable packaging data sheet number (see 5.1).

Custodian:

Army - MU

Preparing Activity:

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Project No. 6650-A149Y

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